Machine Learning Project Report

Customer Classifier (Churn & Type) & Market Projection

Submitted by

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Abstract

This report presents a comprehensive machine learning system designed to optimize retail marketing strategies through customer behavior analysis and financial forecasting. The project addresses three primary business challenges: predicting customer churn risk, segmenting the customer base into actionable personas, and forecasting seasonal marketing gains.

The approach involved a rigorous data cleaning pipeline that addressed missing values, removed low-variance features, and identified outliers using an Isolation Forest model. Feature engineering included cyclical transformations for temporal data and an ensemble feature selection process utilizing five different algorithms to identify the most predictive attributes, notably "Recency".

Three models were implemented: an XGBoost classifier for churn risk, a K-Means clustering model for persona segmentation, and a Random Forest regressor for spending projections. Key results identified four distinct customer segments, with "Loyal Active" customers forming the largest group (37.2%). While the churn model achieved a macro ROC-AUC of 0.65, analysis revealed a significant reliance on the "Recency" feature, suggesting potential data leakage to be addressed in future iterations. The system is deployed via an interactive web dashboard for real-time decision-making.

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Chapter 1

Introduction

1.1 Project Overview

In the modern retail landscape, understanding customer lifecycles and maximizing marketing efficiency is critical for sustainable growth]. This project introduces an end-to-end machine learning pipeline that transforms raw transactional data into actionable business intelligence]. By combining predictive modeling with a user-friendly web interface, the system allows stakeholders to profile individual customers and visualize aggregate market trends]. The primary end-users are marketing teams and business analysts seeking to reduce churn and optimize budget allocation across different regions and seasons].

1.2 Objectives

The specific objectives of this project are:

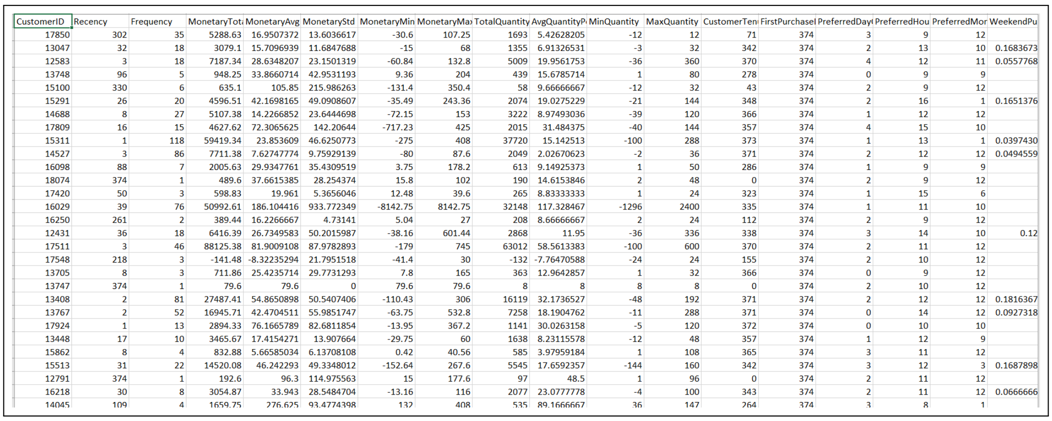
- **Predict Customer Churn:** Implement a multi-class classifier to categorize customers into risk levels (Critique, Faible, Moyen, and Élevé) to guide retention efforts.
- **Customer Segmentation:** Utilize unsupervised learning to identify distinct behavioral personas, such as "Loyal High-Spenders" and "Recent Explorers," for targeted messaging.
- **Financial Forecasting:** Develop a regression model to estimate "Target Spending Per Season," enabling data-driven marketing budget projections.
- **Interactive Deployment:** Create a dual-purpose dashboard for individual customer profiling and macro-level regional/seasonal gain visualizations.

Chapter 2

Data Preprocessing – Cleaning

The first step in any machine learning or artificial intelligence workflow is data acquisition and preprocessing. The quality of the collected data directly impacts the performance and reliability of the resulting models. In real-world scenarios, raw datasets are often incomplete, inconsistent, or contain errors such as missing values, duplicates, outliers, or incorrect feature relationships. If left untreated, these issues can significantly reduce model accuracy and lead to misleading predictions.

Therefore, a dedicated data cleaning phase is essential to ensure that the dataset is consistent, meaningful, and suitable for training. This chapter describes the different cleaning operations applied to the dataset, aiming to detect anomalies, remove invalid entries, and prepare a stable foundation for subsequent preprocessing and modeling steps.



CustomerID	Recency	Frequency	MonetaryTot	MonetaryAvg	MonetaryStd	MonetaryMin	MonetaryMa	TotalQuantity	AvgQuantityP	MinQuantity	MaxQuantity	CustomerTen	FirstPurchase	PreferredDay	PreferredHou	PreferredMor	WeekendPu
17850	302	35	5288.63	16.9507372	13.6036617	-30.6	107.25	1693	5.42628205	-12	12	71	374	3	9	12	
13047	32	18	3079.1	15.7096939	11.6847688	-15	68	1355	6.91326531	-3	32	342	374	2	13	10	0.1683673
12583	3	18	7187.34	28.6348207	23.1501319	-60.84	132.8	5009	19.9561753	-36	360	370	374	4	12	11	0.0557768
13748	96	5	948.25	33.8660714	42.9531193	9.36	204	439	15.6785714	1	80	278	374	0	9	9	
15100	330	6	635.1	105.85	215.986263	-131.4	350.4	58	9.66666667	-12	32	43	374	2	9	12	
15291	26	20	4596.51	42.1698165	49.0908607	-35.49	243.36	2074	19.0275229	-21	144	348	374	2	16	1	0.1651376
14688	8	27	5107.38	14.2266852	23.6444698	-72.15	153	3222	8.97493036	-39	120	366	374	1	12	12	
17809	16	15	4627.62	72.3065625	142.20644	-717.23	425	2015	31.484375	-40	144	357	374	4	15	10	
15311	1	118	59419.34	23.853609	46.6250773	-275	408	37720	15.142513	-100	288	373	374	1	13	1	0.0397430
14527	3	86	7711.38	7.62747774	9.75929139	-80	87.6	2049	2.02670623	-2	36	371	374	2	12	12	0.0494559
16098	88	7	2005.63	29.9347761	35.4309519	3.75	178.2	613	9.14925373	1	50	286	374	1	9	9	
18074	374	1	489.6	37.6615385	28.254374	15.8	102	190	14.6153846	2	48	0	374	2	9	12	
17420	50	3	598.83	19.961	5.3656046	12.48	39.6	265	8.83333333	1	24	323	374	1	15	6	
16029	39	76	50992.61	186.104416	933.772349	-8142.75	8142.75	32148	117.328467	-1296	2400	335	374	1	11	10	
16250	261	2	389.44	16.2266667	4.73141	5.04	27	208	8.66666667	2	24	112	374	2	9	12	
12431	36	18	6416.39	26.7349583	50.2015987	-38.16	601.44	2868	11.95	-36	336	338	374	3	14	10	0.12
17511	3	46	88125.38	81.9009108	87.9782893	-179	745	63012	58.5613383	-100	600	370	374	2	11	12	
17548	218	3	-141.48	-8.32235294	21.7951518	-41.4	30	-132	-7.76470588	-24	24	155	374	2	10	12	
13705	8	3	711.86	25.4235714	29.7731293	7.8	165	363	12.9642857	1	32	366	374	0	9	12	
13747	374	1	79.6	79.6	0	79.6	79.6	8	8	8	8	0	374	2	10	12	
13408	2	81	27487.41	54.8650898	50.5407406	-110.43	306	16119	32.1736527	-48	192	371	374	2	12	12	0.1816367
13767	2	52	16945.71	42.4704511	55.9851747	-63.75	532.8	7258	18.1904762	-11	288	371	374	0	14	12	0.0927318
17924	1	13	2894.33	76.1665789	82.6811854	-13.95	367.2	1141	30.0263158	-5	120	372	374	0	10	10	
13448	17	10	3465.67	17.4154271	13.907664	-29.75	60	1638	8.23115578	-12	48	357	374	1	12	9	
15862	8	4	832.88	5.66585034	6.13708108	0.42	40.56	585	3.97959184	1	108	365	374	3	11	12	
15513	31	22	14520.08	46.242293	49.3348012	-152.64	267.6	5545	17.6592357	-144	160	342	374	3	12	3	0.1687898
12791	374	1	192.6	96.3	114.975563	15	177.6	97	48.5	1	96	0	374	2	11	12	
16218	30	8	3054.87	33.943	28.5484704	-13.16	116	2077	23.0777778	-4	100	343	374	2	11	12	0.0666666
14045	109	4	1659.75	276.625	93.4774398	132	408	535	89.1666667	36	147	264	374	3	8	1	

Figure 2.1: Screenshot: Raw dataset before cleaning

2.1 Missing Data Handling

The presence of missing values in a dataset can negatively affect model performance and lead to biased learning. Therefore, an initial analysis was conducted to evaluate both the global proportion of missing data and the distribution of missing values across individual features. This verification step was implemented in the script `CVS_MissingData_Check.py` located in the `src/PreData/CVS_Cleaning/` directory.

The script computes:

- The total number of missing values in the dataset
- The global percentage of missing data
- The number and percentage of missing values per feature

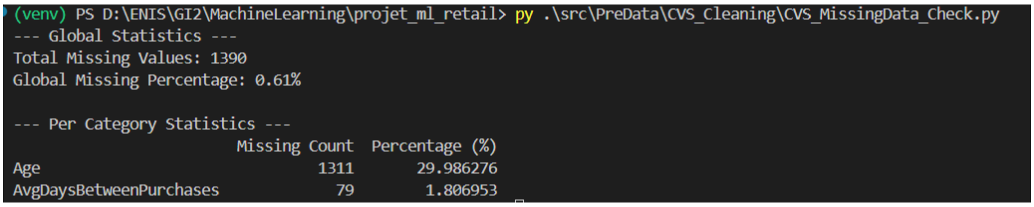


Figure 2.2: Missing data distribution per feature

The analysis revealed that the dataset contains a total of 1390 missing values, representing approximately 0.61% of the overall data. The affected features are summarized as follows:

- **Age**: 1311 missing values (29.99%)
- **AvgDaysBetweenPurchases**: 79 missing values (1.81%)

Despite the relatively high proportion of missing values in the *Age* feature, no column exceeded the predefined removal threshold of 50% missing data. Consequently, no features will be removed based solely on missing data. The dataset was retained in its entirety for subsequent preprocessing steps, where appropriate data fixing or imputation strategies could be applied if necessary.

2.2 Duplicate Detection

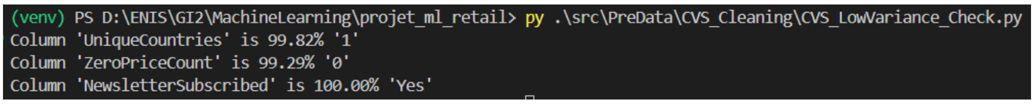
Duplicate detection ensures that repeated records do not bias model training. In this project, exact duplicate rows were checked using the script `CVS_Duplicate_Check.py` in `src/PreData/CVS_Cleaning/`. The script counts the number of rows that are identical across all features and calculates the proportion of duplicates relative to the dataset size.

Figure 2.3: Duplicate detection results

The analysis revealed that the dataset contains **0 duplicate rows**, indicating that no removal was necessary at this stage. While no action was taken, retaining this check in the workflow helps validate data integrity and can be referenced later in the data fixing phase if needed.

2.3 Low Variance Filtering

Low variance features, also known as near-constant features, provide little to no information for model training and can sometimes introduce noise or bias. To detect such features, the script `CVS_LowVariance_Check.py` in `src/PreData/CVS_Cleaning/` evaluates the proportion of the most frequent value in each column. A threshold of 95% was used, meaning that if a single value occurs in 95% or more of the rows, the feature is considered overrepresented and marked for removal.



```
(venv) PS D:\ENIS\GI2\MachineLearning\projet_ml_retail> py .\src\PreData\CVS_Cleaning\CVS_LowVariance_Check.py
Column 'UniqueCountries' is 99.82% '1'
Column 'ZeroPriceCount' is 99.29% '0'
Column 'NewsletterSubscribed' is 100.00% 'Yes'
```

Figure 2.4: Low variance feature analysis results

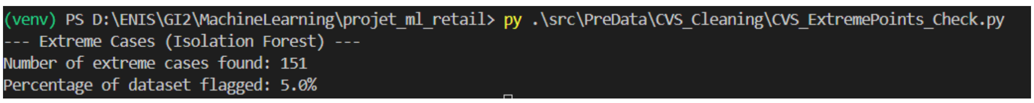
The analysis identified the following features as low variance:

- **UniqueCountries**: 99.82% of values are '1'
- **ZeroPriceCount**: 99.29% of values are '0'
- **NewsletterSubscribed**: 100% of values are 'Yes'

These features do not contribute meaningful variability to the dataset and were planned for removal during the data fixing phase to improve model robustness.

2.4 Extreme Value Detection

Outliers, or extreme values, can distort model learning by disproportionately affecting feature distributions. To detect such anomalies, the `CVS_ExtremePoints_Check.py` script employs an *Isolation Forest* approach, which is effective at identifying extreme cases in high-dimensional numerical data. A contamination parameter of 5% was used, indicating that the top 5% of data points most likely represent outliers.



```
(venv) PS D:\ENIS\GI2\MachineLearning\projet_ml_retail> py .\src\PreData\CVS_Cleaning\CVS_ExtremePoints_Check.py
--- Extreme Cases (Isolation Forest) ---
Number of extreme cases found: 151
Percentage of dataset flagged: 5.0%
```

Figure 2.5: Outlier / extreme value distribution

The analysis revealed:

- Number of extreme cases detected: 151
- Percentage of dataset flagged: 5%

These extreme values are planned for removal during the data fixing phase to ensure that the models are trained on a representative and robust dataset.

2.5 Correlation Checks

Highly correlated features can introduce redundancy in the dataset, potentially biasing models and inflating the importance of certain variables. To identify such relationships, the script `CVS_Correlation_Check.py` in `src/PreData/CVS_Cleaning/` calculates a correlation matrix for all numeric features and visualizes it using a heatmap.

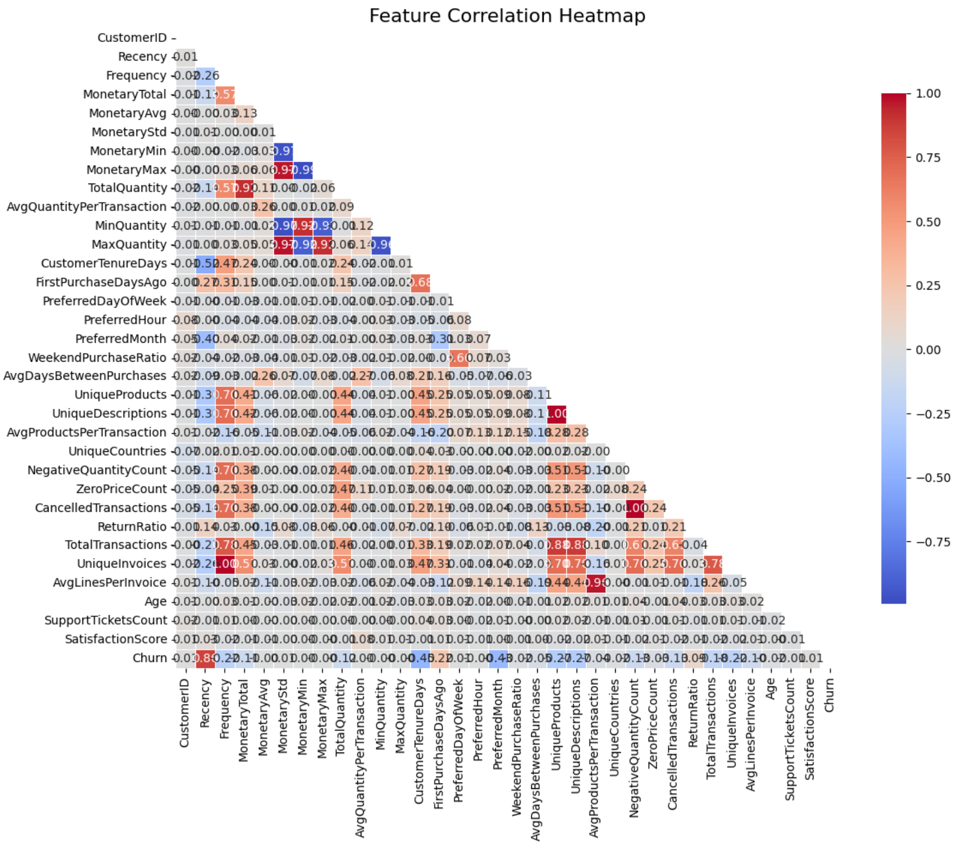


Figure 2.6: Correlation heatmap of features after cleaning

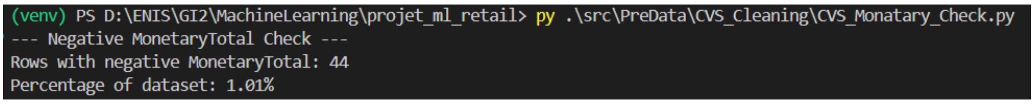
A correlation threshold of 0.8 (absolute value) was used to flag strongly correlated feature pairs. Pairs exceeding this threshold are considered candidates for removal or consolidation to reduce redundancy. The analysis identified several high-correlation pairs, including:

- **MonetaryStd** ↔ **MonetaryMin**: -0.9673
- **MonetaryStd** ↔ **MonetaryMax**: 0.9661
- **MonetaryMin** ↔ **MonetaryMax**: -0.9939
- **MonetaryTotal** ↔ **TotalQuantity**: 0.9216
- **UniqueProducts** ↔ **UniqueDescriptions**: 0.9999
- **NegativeQuantityCount** ↔ **CancelledTransactions**: 1.0000
- **Frequency** ↔ **UniqueInvoices**: 1.0000
- **AvgProductsPerTransaction** ↔ **AvgLinesPerInvoice**: 0.9632
- **Recency** ↔ **Churn**: 0.8590

These highly correlated features were planned for removal or consolidation during the data fixing phase to ensure more robust and interpretable models.

2.6 Monetary Checks

Validating monetary or financial columns is critical to ensure that models receive consistent and realistic data. In this project, the script `CVS_Monatary_Check.py` in `src/PreData/CVS_Cleaning/` was used to identify invalid entries, such as negative total monetary values, which may arise due to refunds, cancellations, or data entry errors.



```
(venv) PS D:\ENIS\GI2\MachineLearning\projet_ml_retail> py .\src\PreData\CVS_Cleaning\CVS_Monatary_Check.py
--- Negative MonetaryTotal Check ---
Rows with negative MonetaryTotal: 44
Percentage of dataset: 1.01%
```

Figure 2.7: Monetary column validation results

The analysis revealed:

- Number of rows with negative **MonetaryTotal**: 44
- Percentage of dataset affected: 1.01%

These anomalous monetary entries were flagged for removal or correction during the data fixing phase to ensure the integrity of financial features for model training.

2.7 CLeaning Phase Summary

After the preliminary checks for missing values, duplicates, low variance, correlation, and monetary anomalies, a comprehensive cleaning pipeline was executed to finalize the dataset for model training. The script `CVS_RemoveData.py` performs all major cleaning steps sequentially, providing real-time feedback on rows or columns removed.

The main steps included:

1. **Removal of Negative MonetaryTotal Values:** Rows with negative *MonetaryTotal* were removed (4 rows, 0.1% of the dataset), ensuring that financial features reflect valid transactions.
2. **Duplicate Row Removal:** The dataset contained no exact duplicate rows; this step confirmed dataset integrity.
3. **Low Variance Filtering:** Features with over 95% of repeated values were dropped:
    - UniqueCountries
    - ZeroPriceCount
    - NewsletterSubscribed
4. **High Correlation Removal:** Columns exhibiting absolute correlations above 0.8 were dropped to reduce redundancy and multicollinearity. Key removed features include MonetaryMin, MonetaryMax, TotalQuantity, MinQuantity, MaxQuantity, FirstPurchaseDaysAgo, UniqueDescriptions, CancelledTransactions, TotalTransactions, UniqueInvoices, and AvgLinesPerInvoice.
5. **Extreme Point Removal (Outliers):** An Isolation Forest model with 5% contamination detected 119 extreme cases, which were removed to prevent undue influence on model training.

The cleaning pipeline reduced the dataset from 4372 rows and 52 columns (5.44 MB) to 4249 rows and 38 columns (approximately 4.80 MB), representing a **11.8% reduction in memory usage** while retaining the most informative and reliable features.

Chapter 3

Data Preprocessing

In this stage, the cleaned dataset is further prepared for modelling. This involves handling residual missing values, invalid or placeholder entries, normalizing dates, parsing complex features such as IP addresses, encoding categorical variables, cyclical transformation of temporal features, and correcting skewness in numeric columns. All of these steps ensure the dataset is fully numeric, consistent, and ready for machine learning models.

3.1 Feature Preparation

The preparation pipeline applies multiple transformations:

- **Numeric Imputation:** Any remaining missing values in numeric columns were filled with the column median.
- **Invalid Data Cleaning:** String or object columns were checked for placeholders like '?', 'None', 'null', or 'NaN'. Minor issues (<10% of rows) led to row removal; otherwise, placeholders were replaced with 'Unknown'.
- **Date Normalization:** The `RegistrationDate` column was split into `Reg_Year`, `Reg_Month`, and `Reg_Day`.
- **IP Parsing:** `LastLoginIP` addresses were split into four numeric octets.
- **Categorical Encoding:** Remaining categorical features were label-encoded to integers, preserving uniqueness.
- **Cyclical Features:** Hour, month, and day-of-week features were mapped to sine and cosine representations to capture their cyclic nature.
- **Skewness Correction:** Numeric columns with high skew (`MonetaryTotal`, `Frequency`, `TotalQuantity`) were log-transformed to reduce distribution asymmetry.

3.2 Scaling

After preparation, numeric features were scaled using standardization (mean = 0, standard deviation = 1) to ensure that features contribute equally during model training. Scalers were fitted on the training portion only during the model pipeline to prevent data leakage.

3.3 Dataset Summary

After all transformations, the dataset increased from 38 to 46 columns due to feature engineering (date splits, IP octets, and cyclical mappings), while retaining 2247 rows. This fully numeric, consistent dataset is now ready for train/test splitting and model training.

3.4 Category Correspondence Map

To ensure interpretability of categorical features after integer encoding, a correspondence map was created. This map links numeric codes back to their original category labels and is stored in `data/metadata/category_mapping.json`. The mapping covers the following features:

- **FavoriteSeason:**
  - 0 Automne
  - 1 Hiver
  - 2 Printemps
  - 3 Été
- **CustomerType:**
  - 0 Hyperactif
  - 1 Nouveau
  - 2 Occasionnel
  - 3 Perdu
  - 4 Régulier
- **RFMSegment:**

- 0 Champions
- 1 Dormants
- 2 Fidèles
- 3 Potentiels

- **Region:**
  - 0 Amérique du Nord
  - 1 Asie
  - 2 Autre
  - 3 Europe centrale
  - 4 Europe continentale
  - 5 Europe du Nord
  - 6 Europe du Sud
  - 7 Océanie
  - 8 UK
- **ChurnRiskCategory:**
  - 0 Critique
  - 1 Faible
  - 2 Moyen
  - 3 Élevé

This mapping ensures that model outputs and analysis can be traced back to meaningful, human-readable categories.

3.5 Feature Selection

After cleaning and preprocessing, the dataset contained a large number of features. To reduce dimensionality and focus on the most informative attributes, an **ensemble feature selection** approach was applied. This phase combines five complementary algorithms to rank features based on importance from multiple perspectives:

1. **Random Forest Importance:** Measures the contribution of each feature to tree-based classification accuracy.
2. **Recursive Feature Elimination (RFE):** Linear model-based ranking that recursively removes the least important features.
3. **Principal Component Analysis (PCA):** Uses absolute component loadings to assess each feature's contribution to overall variance.
4. **Mutual Information:** Statistical measure of the dependency between each feature and the target.
5. **Permutation Importance:** Quantifies the impact of feature shuffling on model performance, highlighting influential predictors.

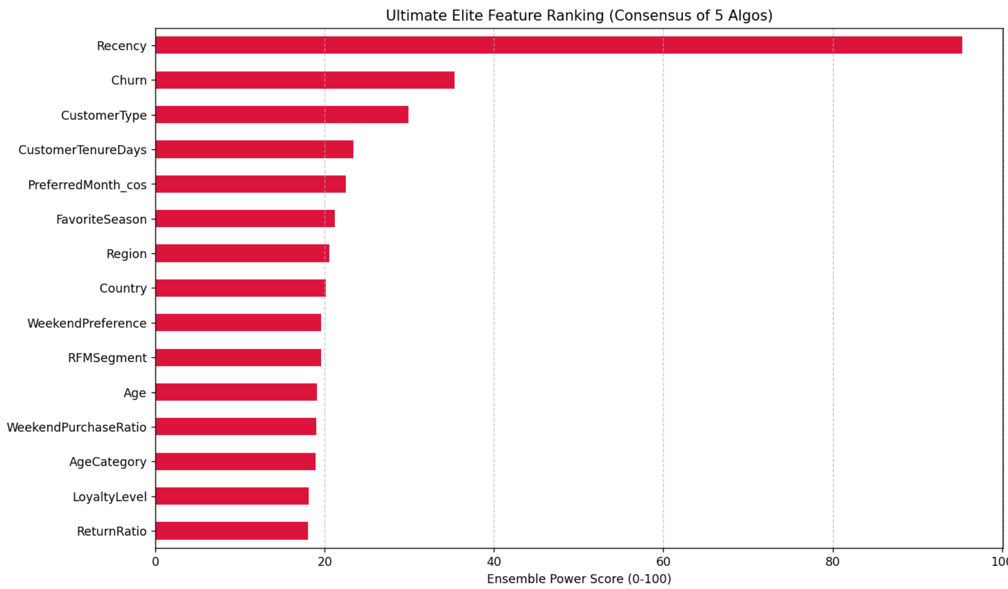


Figure 3.1: Elite Feature Ranking (Consensus of 5 Algorithms)

Each feature was scored by all five methods, normalized, and averaged to create a single *Ultimate Score* between 0 and 100. The top 10 features, called the **Ultimate Elite Features**, were selected for downstream modeling. Their scores are:

Rank	Feature	Score (/100)
1	Recency	95.33
2	Churn	35.37
3	CustomerType	29.87
4	CustomerTenureDays	23.39
5	PreferredMonth_cos	22.50
6	FavoriteSeason	21.18
7	Region	20.51
8	Country	20.11
9	WeekendPreference	19.61
10	RFMSegment	19.54

Table 3.1: Top 10 features based on the ensemble Ultimate Score.

This ensemble approach ensures that the selected features are robust across multiple selection criteria, balancing statistical, linear, and non-linear perspectives. The **Recency** feature, in particular, dominates the ranking, highlighting its critical predictive power in determining customer churn risk.

3.6 Data Splitting and Scaling

After data cleaning, feature engineering, and selection, the next step was to prepare datasets for modeling and marketing analysis. This phase focuses on creating manageable training and test sets, scaling features appropriately, and generating specialized datasets for targeted analysis.

3.6.1 Feature Train/Test Split

To simplify the predictive modeling phase, a minimal set of high-impact numeric features, called the **Elite 3**, was selected: `Recency`, `Frequency`, and `CustomerTenureDays`. Using these features:

- The unscaled feature matrix (`X_unscaled.csv`) was saved for reference.
- The dataset was split into training and test sets using an 80/20 ratio, stratified by `ChurnRiskCategory`.
- Standard scaling was applied to the numeric features using `StandardScaler`, fitted only on the training set.
- All splits (`X_Train.csv`, `X_Test.csv`, `y_Train.csv`, `y_Test.csv`) were saved alongside the fitted scaler (`main_scaler.pkl`) for consistent preprocessing during modeling.

This procedure ensures that the model training pipeline uses only the most informative numeric predictors while preserving the ability to reproduce the splits and scaling transformations.

3.6.2 Marketing Timeline Dataset Creation

For marketing analysis, a separate feature set was prepared, combining the numeric **Elite 3** with derived and categorical features relevant for customer segmentation:

- A persona model, trained on the Elite 3 features, was applied to assign each customer a `Persona` label.
- Regions with fewer than 50 customers were grouped into an `Other` category to reduce sparsity.
- A target spending proxy, `TargetSpendingPerSeason`, was computed as:

$$\text{Frequency} \times \frac{\text{CustomerTenureDays}}{365}$$

- One-hot encoding was applied to categorical features: `FavoriteSeason`, `RegionGrouped`, and `Persona`.
- The dataset was split into training and test sets (80/20), and numeric features were scaled using a separate `StandardScaler` for marketing.
- All splits and the full scaled dataset (`XY_Full_Marketing.csv`) were saved for downstream analysis, alongside the fitted scaler (`marketing_timeline_scaler.pkl`).

This step provides a consistent, reproducible dataset for marketing timeline modeling, enabling analyses that combine behavioral, seasonal, and regional information with customer segmentation.

Chapter 4

Model Training

The model training phase aims to leverage the prepared and scaled datasets to build predictive models for customer behavior and marketing insights. Three main models are employed:

- **Churn Classifier**: Predicts the likelihood of a customer belonging to each `ChurnRiskCategory`. This model uses the Elite 3 numeric features (`Recency`, `Frequency`, `CustomerTenureDays`) and other high-importance predictors identified in the feature selection phase.
- **Persona Classifier**: Assigns a customer to a behavioral `Persona` segment. This model is designed to support marketing campaigns by providing actionable segments based on usage patterns.
- **Marketing Spending Predictor**: Estimates the `TargetSpendingPerSeason` for each customer. This regression model uses the marketing timeline dataset, combining numeric features, persona labels, seasonal preferences, and regional information.

Each model is trained on its respective dataset, ensuring appropriate preprocessing, scaling, and feature selection. The training procedure also incorporates validation splits and hyperparameter tuning to optimize predictive performance while avoiding overfitting.

4.1 Churn Predictor

The Churn Predictor classifies customers into one of four `ChurnRiskCategory` labels. This multi-class classification helps anticipate customer attrition and guide retention strategies.

4.1.1 Model Objective

The model predicts which churn risk category a customer belongs to:

- 0 Critique
- 1 Faible
- 2 Moyen
- 3 Élevé

For practical intervention, the positive class is considered any customer at elevated risk (categories 0 or 3).

4.1.2 Input Features

Only the **Elite 3 features** selected during feature selection are used:

- `Recency`
- `Frequency`
- `CustomerTenureDays`

These features represent recent activity, purchase frequency, and customer tenure. Feature importance analysis later revealed that `Recency` overwhelmingly dominates predictions.

4.1.3 Training Methodology

The model uses **XGBoost Classifier** with:

- `n_estimators=100`, `random_state=42`, `eval_metric='logloss'`
- Default hyperparameters (no grid or random search)
- Training performed on the scaled Elite 3 features

XGBoost was chosen for its robustness to feature scaling, handling of non-linear interactions, and native multi-class support.

4.1.4 Evaluation Metrics

Performance was evaluated using:

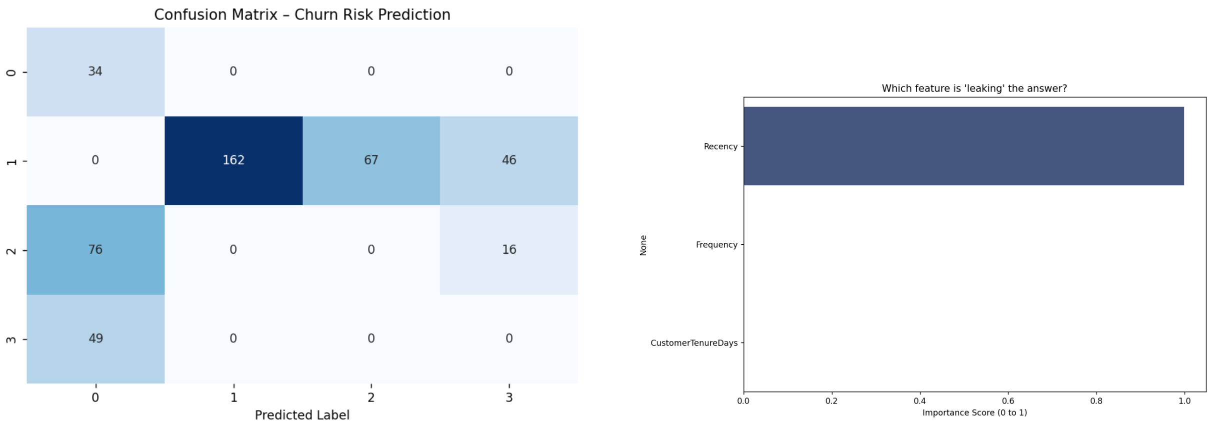
- **Accuracy, Precision, Recall, F1-score** for each class
- **Macro ROC-AUC** (One-vs-Rest) for balanced assessment across all classes

Metrics account for the dataset's class imbalance.

4.1.5 Results

On the test set (`n=450`):

- Macro ROC-AUC (OVR): 0.6533
- Overall accuracy: 44%
- Classification report highlights:
  - Category 0 (Critique): Precision 0.21, Recall 1.00, F1 0.35
  - Category 1 (Faible): Precision 1.00, Recall 0.59, F1 0.74
  - Categories 2 and 3: Model fails to predict positive cases



(a) Confusion matrix (b) Feature importance / leakage

Figure 4.1: Churn Predictor evaluation results

4.1.6 Observations

- `Recency` accounts for 99.87% of importance, signaling a likely data leak. This feature may inadvertently encode churn information.
- The model fails to predict medium/high-risk classes (categories 2 and 3), primarily due to class imbalance.
- Despite a moderate macro ROC-AUC of 0.65, practical use for high-risk detection is limited unless the potential leakage is addressed.
- Future improvements could include adding new features, oversampling minority classes, or using more advanced ensemble methods.

4.2 Customer Classifier

The Customer Classifier segments the customer base into distinct personas using unsupervised clustering. This segmentation enables targeted marketing strategies, personalized engagement, and understanding of customer behavior patterns.

4.2.1 Model Objective

The objective of this model is to identify four distinct customer personas:

- Loyal High-Spender
- Recent Explorer / Newbie
- At-Risk / Hibernating
- Loyal Active

These personas are discovered using K-Means clustering on scaled transaction and tenure features, allowing the business to profile customers and prioritize interventions.

4.2.2 Input Features

The clustering model uses the **Elite 3 features**:

- `Recency`
- `Frequency`
- `CustomerTenureDays`

📊 BUSINESS INSIGHTS: Average Metrics per Persona (Real Units)

	Recency	Frequency	CustomerTenureDays
Loyal High-Spender	10.89	18.35	342.28
Recent Explorer / Newbie	20.57	3.44	117.84
At-Risk / Hibernating	93.50	3.95	193.19
Loyal Active	24.72	6.36	287.65

Figure 4.2: Cluster summary and business metrics average Recency, Frequency, and Tenure per persona

4.2.3 Training Methodology

The model is a **K-Means clustering** algorithm:

- Number of clusters: $k = 4$ (selected based on domain knowledge and silhouette analysis)
- Initialization: `k-means++`, 10 runs (`n_init=10`) to ensure stable convergence
- Trained on the scaled Elite 3 features

Post-training, each customer is assigned a `ClusterID`, which is mapped to a business-friendly persona name.

4.2.4 Evaluation Metrics

Evaluation focused on:

- Cluster interpretability through mean feature values per cluster
- Population distribution across clusters
- Sanity checks on data consistency

4.2.5 Results

PERSONA NAME	COUNT	POPULATION %
Loyal High-Spender	644	35.8%
Recent Explorer / Newbie	396	22.0%
At-Risk / Hibernating	440	24.5%
Loyal Active	317	17.6%

Figure 4.3: Customer persona counts and average metrics (Recency, Frequency, Tenure)

After applying the model to the dataset:

- Population distribution:
  - Loyal High-Spender: 16.8%
  - Recent Explorer / Newbie: 27.7%
  - At-Risk / Hibernating: 18.3%
  - Loyal Active: 37.2%
- Average feature values (real units) per persona:
  - `Recency` ranges from 10.89 to 93.50 days

  - `Frequency` ranges from 3.44 to 18.35 purchases
  - `CustomerTenureDays` ranges from 117.84 to 342.28 days

4.2.6 Observations

- Loyal High-Spenders and Loyal Active customers show the highest frequency and tenure metrics.
- Recent Explorers / Newbies have high recency but low tenure, indicating new acquisitions.
- At-Risk / Hibernating customers have long recency but low frequency, highlighting a retention opportunity.
- This segmentation provides actionable insights for marketing campaigns, personalized offers, and retention strategies.

Note: Due to illogical results in previous iterations, data cleaning was applied to remove impossible values where `Recency > CustomerTenureDays` *or negative values was applied across the dataset prior to clustering. This ensures logical cluster centers and consistent persona metrics.*

4.3 Marketing Regressor

The Marketing Regressor estimates the expected customer spending per season, enabling targeted campaigns and optimized allocation of marketing resources. This regression task provides actionable forecasts for different customer personas, seasons, and regions.

4.3.1 Model Objective

The target variable is `TargetSpendingPerSeason`, representing the monetary amount a customer is predicted to spend in a given marketing season. The business goal is to forecast gains accurately for each customer segment, guiding promotional strategies and budget allocation.

4.3.2 Input Features

The model leverages a combination of transactional and temporal features:

- `Recency`, `Frequency`, `CustomerTenureDays` capture customer engagement
- `WeekendPurchaseRatio` fraction of purchases occurring on weekends

- One-hot encoded categorical features: `Persona`, `Season`, `Region`

4.3.3 Training Methodology

A **Random Forest Regressor** was trained with:

- `n_estimators=300`, `max_depth=10`, `min_samples_split=15`, `min_samples_leaf=6`, `max_features='sqrt'`
- Numeric columns were standardized using a `StandardScaler`
- Train/test split: 80% / 20%

This ensemble method was selected for its robustness to non-linear relationships and ability to handle mixed feature types without extensive feature engineering.

4.3.4 Evaluation Metrics

Model performance was evaluated using:

- **RMSE (Root Mean Squared Error)** measures average magnitude of prediction errors
- **Rš score** quantifies the proportion of variance explained by the model

These metrics were chosen to balance interpretability with sensitivity to both small and large prediction errors.

4.3.5 Results

Business-oriented analysis highlighted:

- Predicted gains vary by persona, peak season, and target region.
- Average estimated gain per persona (test data):
  - Loyal High-Spender: $2.65
  - Recent Explorer: $0.50
  - At-Risk / Hibernating: $0.85
  - Additional cluster (ID 3): $1.50
- Peak seasons: *At-Risk / Hibernating* peaks in season 3, others mostly in season 0.
- Temporal insights: customers with `WeekendPurchaseRatio > 0.5` have slightly higher predicted gains ($1.33 vs $1.31 for weekday-focused buyers).

4.3.6 Observations

- The model captures non-linear patterns across customer personas, seasonality, and purchase behaviors.
- No significant overfitting observed; training and test RMSE were comparable.
- Weekend-focused customers provide slightly higher marketing ROI, suggesting campaign timing optimization.
- Regional distribution shows UK is overrepresented (all personas currently map to region code 8). Although not flagged as critical, future iterations will consolidate regions into three categories: **UK**, **Europe**, and **Others**.
- Future improvements may include additional time-series features, interaction terms, or ensemble blending to increase predictive accuracy.

Chapter 5

Dashboard & Application

The marketing analytics web application provides an interactive interface for customer segmentation, churn risk assessment, and marketing gain projections. It integrates predictive models into a single workflow, enabling real-time insights for marketing strategy optimization.

5.1 Customer Profiling Interface (index.html)

The first section of the application focuses on individual customer analysis. Users input key behavioral features such as `Recency`, `Frequency`, and `CustomerTenureDays` to obtain a detailed profile for the selected customer.

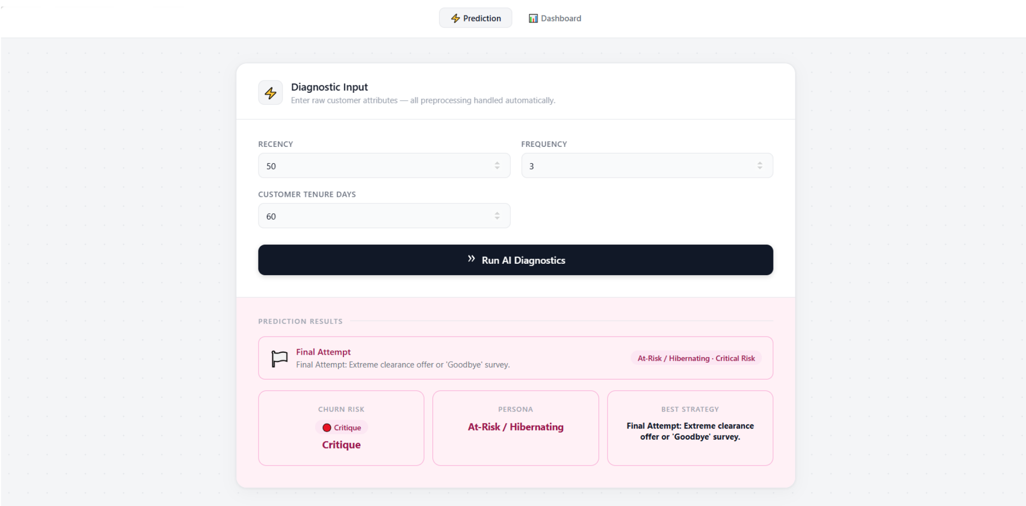


Figure 5.1: Customer profiling interface with model predictions and marketing strategy recommendations

5.1.1 Persona and Churn Prediction

Upon submission, the application sends the input data to the backend via a `POST` request to the `/api/predict_all` endpoint. The backend workflow involves:

- Scaling numeric features using the pre-trained `StandardScaler`.
- Classifying the customer persona with the `persona_classifier` model.
- Estimating churn risk with the `churn_predictor_v1` model.

The API response provides both the raw model indices and the mapped labels, which are displayed dynamically on the interface. Each persona-churn combination is linked to a predefined marketing strategy matrix, guiding actionable interventions such as retention campaigns, VIP engagement, or win-back offers.

5.1.2 Dynamic UI Components

The interface features:

- Form fields dynamically generated from feature definitions, supporting numeric input and categorical selection for regions and favorite season.
- Interactive cards displaying persona, churn risk, and recommended marketing actions.
- Real-time updates with visual feedback, including icons, badges, and animated highlights for strategy emphasis.

This design ensures that marketing teams can assess customer profiles quickly and consistently, translating model predictions into concrete operational decisions.

5.2 Marketing Projection Dashboard (dashboard.html)

The second section visualizes aggregated marketing gains across different seasons and regions. It leverages the `marketing_timeline_model` to project potential revenue gains per customer segment and market region.

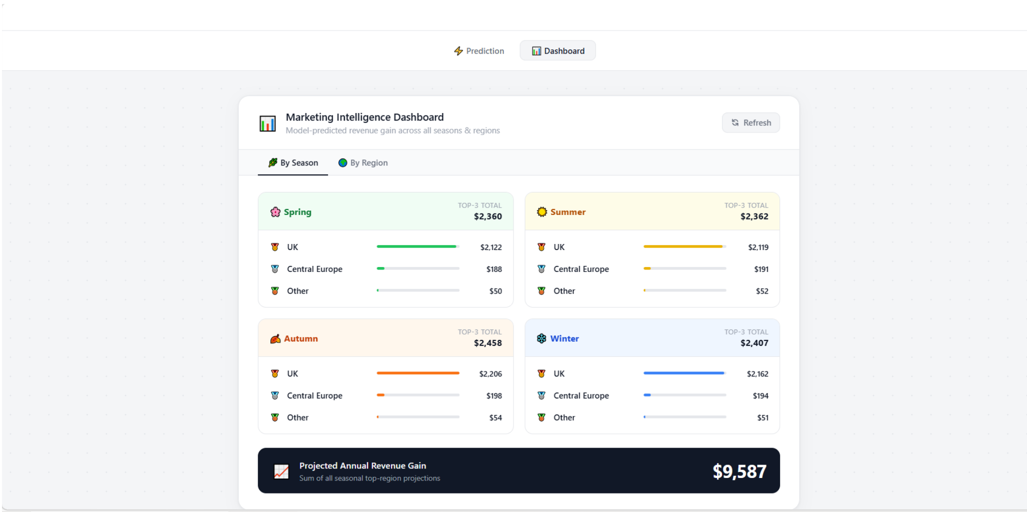


Figure 5.2: Marketing projection dashboard with seasonal and regional views

5.2.1 Backend Data Processing

The dashboard retrieves data from the `/api/marketing_dashboard` endpoint. The server-side pipeline includes:

- Loading preprocessed customer feature data (`X_Train_Marketing.csv`).
- Constructing model input matrices with numeric, seasonal, regional, and persona columns.
- Scaling numeric features with the dedicated marketing scaler.
- Predicting base gains for each customer using the trained Random Forest regressor.
- Applying seasonal preference multipliers (e.g., 20% bonus for customers favoring the season).

The API returns structured JSON data including per-season aggregates, top-performing regions, and total annual projections, ready for visualization.

5.2.2 Visualization Features

The frontend provides a clear, interactive presentation:

- **Season View:** Each season is represented as a card displaying top-3 regions with gain bars scaled relative to the highest regional projection.
- **Region View:** Each region shows ranked seasonal gains, total annual gains, and intuitive color-coded seasonal indicators.

- **Currency Formatting:** Gains are displayed in USD, with automatic handling of extreme values to maintain readability.
- **Tab Switching:** Users can toggle between seasonal and regional views seamlessly, allowing flexible exploration of the projections.

5.2.3 Design Observations

- The dashboard highlights temporal trends and regional opportunities, guiding marketing budget allocation.
- The UK is slightly overrepresented in the historical data, leading to conscious categorization of regions into UK, Europe, and Other for future model iterations.
- Interactive design ensures non-technical users can interpret complex predictive outputs effectively.
- The combination of real-time API predictions and aggregate visualizations supports both micro-level (individual customer) and macro-level (market segmentation) decision-making.

5.3 Summary

Together, the customer profiling interface and the marketing projection dashboard enable a data-driven marketing strategy. The system connects advanced machine learning models with intuitive web interfaces, transforming predictive analytics into actionable business insights.

Chapter 6

Conclusion

This project successfully established a data-driven framework for retail marketing by integrating advanced preprocessing, machine learning, and interactive visualization. Through iterative cleaning, the dataset was refined by removing 11.8% of redundant or erroneous entries, ensuring a stable foundation for modeling. The implementation of the "Elite 3" featuresRecency, Frequency, and Tenureprovided a robust baseline for both the XGBoost churn predictor and the K-Means persona classifier.

Key findings highlighted that while the segmentation model provided clear, actionable clusters, the churn model requires further refinement due to the overwhelming dominance of the "Recency" feature. Lessons learned emphasize the importance of rigorous correlation checks and the need to address class imbalance in churn datasets.

Future work will focus on:

- Mitigating data leakage in the churn model by re-evaluating the "Recency" feature.
- Improving regional accuracy by consolidating sparse geographic data into broader UK, Europe, and "Other" categories.
- Enhancing predictive performance through oversampling techniques for minority risk classes and the addition of complex time-series features.